numbt:. pUlic'Jluly bdo"'· 'peed. of 2 mI.tc.
:-40N.nETURN VALVE
C
V
,--+ ----· ----· ~4· _!_ '.
" c - "
It = 0.)
I--_____ -'00ATE VAl.VE
k - /0.0
DISC VA LVE
FICURE 11
FIG URE 9
The co cf!i ~i .nl above i. ~ mean yaJyC and dOcl in f~Cl vary
21 R.yn old'J nY,nlM:, (i.e. ~ ar;':, wilh V (0 : con ll1nl D .n d
~.) u ..... U n wjlh ,clu ivc ,,,.,&-'1,,c.1.
PIPE FRICTION LOSSES
fiGURE I I
k " O . ~
aUT HR F L Y VALVE. _____ __ _ MSC 83/28/Add.2
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Page 8

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Figure 13
Cross-flooding through a series of structural ducts with 1 manhole

1 00313 . 0 2748 . 0 < <+= ii L ifLk
4 17386 . 0 0212 . 1 6873 . 0 0986 . 023 + + = iiii L ifLLLk
434 . 1 >= iL ifk

Note: k is the friction coefficient related to each space between two adjacent girders. k is evaluated with effective cross-section area theref ore in calculations use the real cross-section area A and not S equiv. The pressure loss for entrance in the first manhole is already computed in the calculation, and k = 1 has to be added to take into account the outlet losses.

Li (in metres)
RESOLUTION MSC.245(83)
(adopted on 12 October 2007)

---CHUNK---

SOURCE: MSC.245(83).pdf
in the calculation, and k = 1 has to be added to take into account the outlet losses.

Li (in metres)
RESOLUTION MSC.245(83)

Figure 14
Cross-flooding through a series of structural ducts with 2 manholes

1 00627 . 0 4045 . 0 < <+= ii L ifLk
4 1356 . 0 1401 . 1 3593 . 0 0424 . 023 + = iiii L ifLLLk
417 . 1 >= iL ifk

Note: k is the friction coefficient related to each space between two adjacent girders. k is evaluated with effective cross-section area theref ore in calculations use the real cross-section area A and not S equiv. The pressure loss for entrance in the first manhole is already computed in the calculation, and k = 1 has to be added to take into account the outlet losses.

APPENDIX 3

EXAMPLE USING FIGURES FOR A PASSENGER SHIP

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RECOMMENDATION ON A STANDARD METHOD FOR
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APPENDIX 3

EXAMPLE USING FIGURES FOR A PASSENGER SHIP

Dimension of the considered cross-flooding pipe:

Diameter m D39 . 0=

Length m l0 . 21=
Cross-section area 212 . 0m S=
Wall thickness mm t5 . 17=

k-values for the considered cross-flooding system:

Inlet 0.45
Pipe friction D l02 . 0 1.08
2 radius bends (° =45) 0.36
Non-return valve 0.50
Outlet 1.00
= 39 . 3k

Sufficient air venting is assumed to be in place.

From this follows:

=
k
F 1

54 . 0
39 . 3
1 = =F

Time required from commencement of cross flooding o to the final equilibrium condition f :

=
0
0
0
1
1
2
1
2
H
hgH
H
h
F S
WT

f
f
f
f

Head of water before commencement of cross-flooding:

m H3 . 50 =

Volume of water which is used to bring the ship from comme ncement of cross-flooding to the final equilibrium condition:

3365m Wf =
RESOLUTION MSC.245(83)
(adopted on 12 October 2007)
RECOMMENDATION ON A STANDARD METHOD FOR

---CHUNK---

SOURCE: MSC.245(83).pdf
Volume of water which is used to bring the ship from comme ncement of cross-flooding to the final equilibrium condition: